Rich regime

scale of initialisation

Rich regime: $(w_i^0, a_i^0) \sim \frac{\lambda}{\sqrt{m}} D$ → feature learning happens

In the following $\lambda \ll 1$

D such that $\begin{cases} |a_i| \geq \|w_i\| & \text{for any } i \\ \sum_{i=1}^m (a_i^0)^2 \leq \lambda \end{cases}$

→ balanced initialisation

→ finite variance

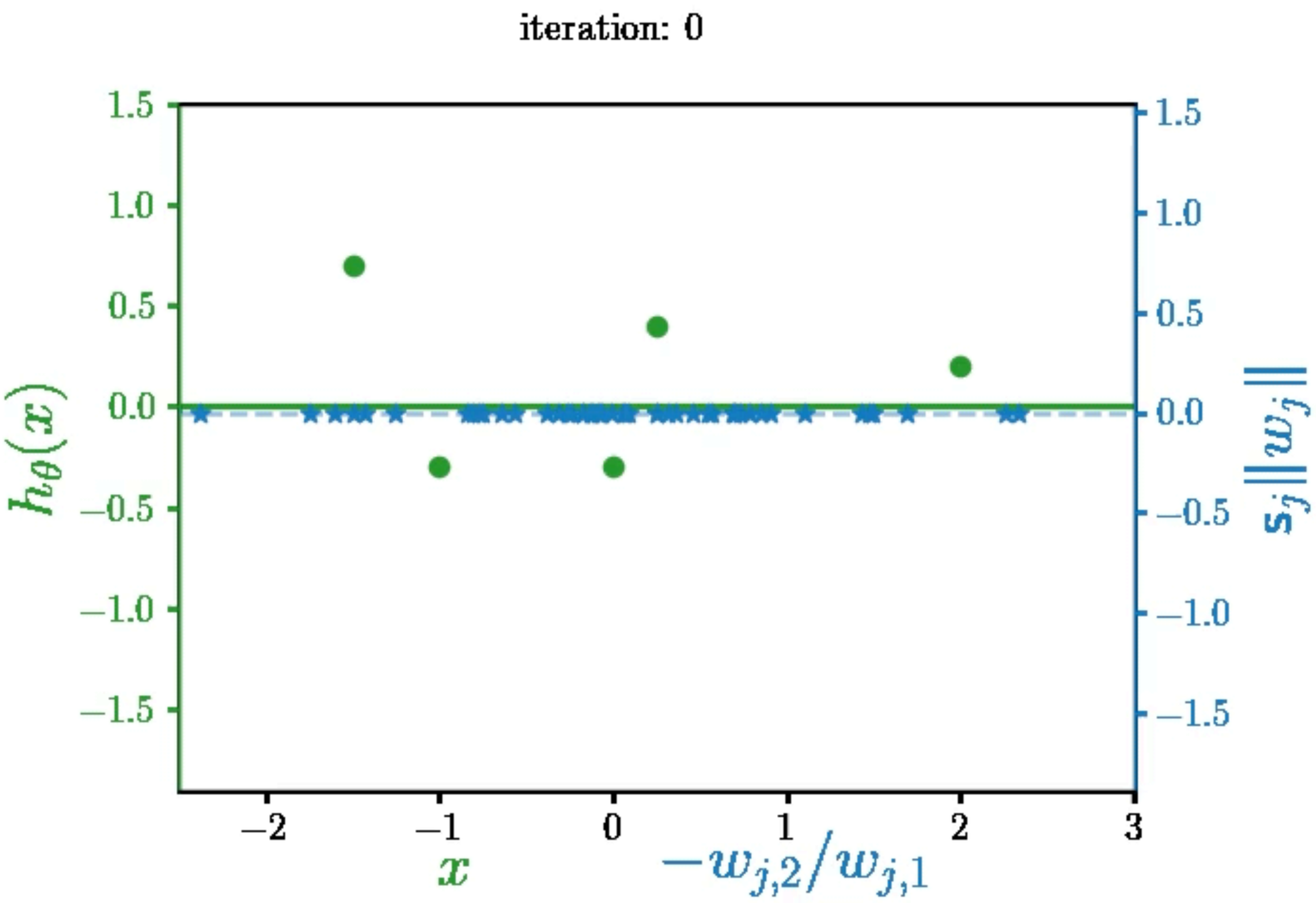
Example: $\begin{cases} w_i \sim \lambda_m \mathcal{U}(B(0,1)) \\ a_i = \pm \lambda_m \end{cases}$

Lemma (balancedness)

$$\boxed{(a_j^t)^2 - \|w_j^t\|^2 = \text{constant}}$$

a_i^t **has a constant sign**

Illustration of rich regime

- slower convergence to 0 loss
- **feature learning**
- **simple** estimator

more explanations to come…